

David Williamson

Oil & Gas Executive | Corporate Strategy (A&D) | Cradle to Grave Development
Geology | Mentor | Board of Directors

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SUMMARY

Repeat entrepreneur, who understands that working capital is needed to survive but quality human capital is essential to thrive. Geoscientist with 30 years' experience working in energy, mining, and emerging technologies. Experience includes both conventional and unconventional plays, with several hundred wells—vertical and horizontal—having reached TD under my watch. A technical leader, with a track record of increasing levels of responsibility, delivering actionable business recommendations and value creation. I thoroughly enjoy the full cycle of prospect execution, from regional studies to proof of concept through the drill bit to economically developing a field. As a successful consultant, adept at relationship building and dovetailing opportunity creators with opportunity seekers. I wish to apply my skills and training to project-oriented, high-yield ventures in an environment whose philosophy matches mine: a commitment to success and a desire to utilize cutting-edge approaches to problem solving.

CORE COMPETENCIES

Data Integration and Analytics – Portfolio/Project Management – Continuous Improvement – Team Leadership

Opportunity Generation – Collaboration – Competitor Analysis – Applied Technology – Coaching/Mentorship

EXPERIENCE

Wheelhouse Geoscience (Frisco, Texas)

2020 – Present

Founder | Chief Executive Officer

- **Founded a multidisciplinary geoscience consultancy** in response to the unprecedented downturn in the petroleum industry, where both public and private companies were forced to offload critical internal expertise owing to capital constraints.
- Wheelhouse Geoscience is a consortium of highly skilled geologists, geophysicists, engineers, geographers, environmental specialists, and IT gurus, who have deep experience and demonstrable accomplishment in their respective disciplines.
- Working individually or deployed in teams, Wheelhouse Geoscience associates and affiliate partners bring an innovative mindset, critical problem-solving skills, and an arsenal of best practices developed over their successful careers.
- In addition to being the principal scientist, **manage subcontractors, negotiate with third-party service providers, and handle all marketing and billing.**
- Since inception, Wheelhouse has successfully completed over 100 buyer-side and seller-side technical evaluations, resulting in **transactions approaching \$600 million.**
- Non-petroleum projects include technical and business development support for gold and rare-earth elements (REE) mining, carbon capture and sequestration (CCS), lithium extraction from petroleum wastewater, green/renewable natural gas (RNG) production from agricultural stock, etc.

HomeWorld Educational Resources (Dallas, Texas)

2022 – Present

Co-Founder | Chief Executive Officer

- **Co-founded a tech startup non-profit organization (NPO)**, with a mission to increase science literacy and STEM education in K-12 classrooms utilizing an AI-created and curated instructional platform and U.S. Department of Education Next Generation Science Standards (NGSS) curricula.
- Talented technical team assembled having expertise in multimedia, deep learning, AI data and analysis, and earth systems science. Content and learning methodology advising through committed relationships with faculty and researchers at 7 U.S. universities.
- **UT Dallas involvement:** 2024 UTD Big Idea Competition semi-winner (the only NPO to reach this stage); 2024 UTD NPO Booster Camp; invited to Phase II of 2025 Tool Competition (out of 3,000 applicants)
- **Awarded 2025 funding through the National Science Foundation (NSF) National Artificial Intelligence Research Resource (NAIRR);** Field of Science: Visualization and Human-Computer Systems.

Yates Industries, LLC (Santa Fe, New Mexico)

2020 – Present

Chief Geoscientist

- In charge of all science for all company assets throughout Texas, New Mexico, Oklahoma, and Kansas. Operational lead for same. **Manage direct reports** (3 geophysicists, 1 engineer) in ongoing assessment and risk analysis of oil and gas development.
- Lead role in 2024 marketing efforts to divest non-core assets in Wyoming and Texas resulting in debt reduction and improved profit margins.
- Principal for all business development activities, including sourcing opportunities, technical and economic analysis, and due diligence. **2025 acquisitions success increased company's daily oil and gas production by 120% and 151%, respectively.**
- After months of preparation and depositions, provided expert testimony under oath to a 3-judge panel arbitration in a 2023 legal matter that settled in favor of Yates Industries.

Monadnock Resources (Addison, Texas)

2015 – 2020

Co-Founder | Vice President, Geoscience

- With founding team, built out comprehensive technical deep-dive and economic modeling for energy startup.
- Fundraising junket included presenting to 9 private equity companies in Houston, Dallas, and New York. **Received term sheets from 3 of them.**
- **Awarded \$100 million in equity backing** from Kayne Anderson, the company focused on the horizontal development of San Andres carbonate reservoirs residing within the NW Shelf of the Permian Basin, West Texas.
- **In charge of all aspects of the company's geoscience**, including play fairway analysis, detailed reservoir studies employing comprehensive petrophysical and core analyses, well planning, geosteering/drilling execution, and assessing bolt-on and new play business development opportunities.
- From concept to execution, **organically grew to ~60,000 operated acres and 2,500 gross barrels of oil per day**, extending the economic play limits of some of the Permian's most prolific legacy fields, such as Wasson Field (>2 billion cumulative barrels produced).
- As a co-founder and managing executive of the company, routinely represented the company's best interests to the Board and external parties, was heavily involved in all strategic planning, and contributed extensively to the company's ongoing risk management.
- **Raised \$320 million** via Kayne Anderson re-investment and Wells Fargo lending to establish Monadnock Resources II, LLC. Won competitive bid to acquire a large (10,000 barrels of oil equivalent per day) conventional, South Texas asset in the first quarter of 2020 (transaction did not close due to COVID-19).

Forge Energy (San Antonio, Texas)

2012 – 2015

Vice President, Geoscience

- Recruited during the company's second year to lend **technical expertise and leadership** to drive growth. With a \$400 million equity commitment from EnCap Investments and Pine Brook Partners, ramped up from 1 drilling rig to running 3 rigs constantly.
- Technical work helped increase daily production from 220 net barrels of oil equivalent per day (boepd) to over 3,500 boepd, **an increase of 1,490%.**
- Supervised all aspects of geoscience, including operations, development, and new ventures. Evaluated multiple prospect data packages and divestiture data rooms.
- As VP, Geoscience, **managed direct reports** and helped Forge take a total petroleum systems approach to play entry and value creation. A creative, technically savvy team, Forge **saw net asset value (NAV) grow from \$100MM to over \$600MM in ~2.5 years** during my tenure with the company.
- Smart execution and disciplined capital outlay created a multi-asset portfolio able to expand through cash flow. **Portfolio sold in 2018 in two transactions totaling over \$1 billion.**

Pioneer Natural Resources (Las Colinas, Texas)

March 2004 – July 2012

Senior Geologist, New Plays and Shale Technology Team, 2011 – 2012

- Worked in a small, dynamic team established to fill a corporate-level organic growth role largely focused on unconventional plays within North America. A multidisciplinary group, we focused on sequence stratigraphy, organic geochemistry,

unconventional petrophysics, seismic volume interpretation and attribute analysis and a material science approach to reservoir engineering and completions.

- Worked primarily on geochemistry studies of potential resource plays in the Permian Basin, East Texas, and onshore Louisiana, as well as our existing Barnett asset area.
- **Mentor or co-mentor to 4 summer intern students.**
- Utilized my previous career experience when acting as sole geologist for the geologic due diligence of multiple frac sand mines belonging to Carmeuse Industrial Sands. My findings were **presented to Management Committee and the Board of Directors.** This **\$297 million acquisition** officially closed on Apr. 2, 2012 and complemented Pioneer's vertical integration initiatives by securing company owned/operated frac sand supplies for decades.

Senior Geologist, Barnett Asset Team, 2009 – 2011

- Team's operations geologist responsible for geosteering horizontal wells in the liquids-rich Combo Play of Montague and Wise Counties (92,000 gross acres). Wellsite geologist as needed for all coring and logging operations. We drilled the company's first Combo Play wells – 52 horizontals and 9 pilots – during my tenure with the team.
- **Production grew to 10.3 Mboepd at the time of its sale, with net cash proceeds of \$155MM.**
- The geologist in charge of 2 joint ventures in Denton and southern Wise counties. Involved: periodic seismic interpretation of geohazards, due diligence of proposed well programs, and conducting partner technical meetings.
- Other duties included mentoring my replacement for operations geology functions; ongoing competitor analysis; and a number of special projects, such as a pilot XRF study of well cutting samples for mineralogy and brittleness comparison, water resource petrophysics, etc. which were presented to the company's entire G&G staff.

Senior Geologist, West Texas Growth Team, 2007 – 2009

- One of 4 original members of a West Texas Growth Team created as the beginning of a long-lived effort to introduce additional geologic pay zones to Pioneer's aggressive drilling program. Regional interpretation of complex seismic geometries was performed on 34 3D surveys and 5,200 line miles of 2D across the basin using Paradigm and Kingdom SMT software platforms.
- The crown jewel of this team's efforts was to lay the foundation for a comprehensive understanding of Wolfberry economic potential in the Midland Basin. Wolfberry production **became the company's premier asset, with production of 700,000 barrels of oil equivalent per day (boepd)** at time Pioneer was acquired by ExxonMobil.
- Beginning in 2009, was the lead geologist for all "shallow" (Guadalupean-aged) reservoirs, where I continued our regional geologic investigations, conducted detailed field studies, and identified and high-graded a portfolio of leads. This peer-reviewed work culminated in a joint venture that tested a San Andres Formation target via a horizontal reentry, the company's first.

Exploitation & Development Geologist, West Texas Asset Team, 2004 – 2007

- Responsible for new drill and recompletion operations in tight oil and gas plays in 6 producing fields in the Val Verde Basin and Eastern Shelf of West Texas. Using 2 drilling rigs and 2 workover rigs, developed assets targeting both carbonate and clastics facies. Based on 2005 drilling results, doubled our Eastern Shelf presence by leasing an additional 20,500 acres. Charged with extending play fairways through mapping, devising and executing a successful infill program, and instituting a twin shallow program to test and exploit uphole potential. **49 wells drilled and completed by the end of 2006.**
- Exploitation and development geologist for the Spraberry Trend field (Midland Basin). When first hired by Pioneer in 2004, was one of only two geologists working the Permian Basin assets. Executed an aggressive drilling program that employed 6 rigs when I first came to the team and **was running 16 rigs (350 wells per year) by the end of my rotation.** While production companywide increased by 12% that year, Spraberry assets outpaced these metrics by **realizing 21% production growth** over the same period.

Martin Marietta Materials (Denton, Texas)

2000 – 2004

Division Geologist, Southwest Division

- Division geologist for the 2nd largest U.S. aggregates producer (\$8B Market Cap). Responsible for the drafting of 15-year mine plans that directed the operations within the **50+ quarries and distribution yards under my care in TX, OK, AR, and LA.**
- Directed all aspects of exploration within active properties and green field sites.
- Generated reserves analyses, determined reserve life for each quarry or prospect and **made recommendations about the commerciality of deposits.**

- In a lead role for all aspects of mine permitting for Oklahoma sites: created all map exhibits and co-authored the permit and bond paperwork, coordinated efforts with the Oklahoma Department of Mines, and **provided expert testimony in public hearings**.
- Managed direct reports: assigned projects and evaluated progress; mentored reports in the geological aspects of the mining industry and trained personnel in the use of Geographic Information Science and GIS software.

University of Texas at Dallas (Richardson, Texas)

1999 – 2000

Teaching Assistant, Department of Geosciences

- TA for the following courses: Field Geology I (Geological Field Techniques); Structural Geology; Field Geology II (Summer Field Camp).
- Enhanced students' analytical and interpretive skills by assisting in the development of local and regional geologic histories and the generation of maps, cross sections, stratigraphic columns, and stereograms, using manual and computer-aided techniques.

ARCO International Oil & Gas Company (Plano, Texas)

1996 - 1999

Geological Technician, New Ventures Exploration, SE Asia Division

- Worked on the research and computer generation of montages and maps, including geologic, seismic base, subsurface structure, and activity maps, primarily using ArcView, CGMAGE, and StratWorks; digitized maps and well logs using NeuraMap and NeuraLog; manually drafted cross sections.
- Engaged in various support duties for numerous farmouts, primarily of multimillion-dollar, multiple partner exploration ventures in Myanmar, China, and the Philippines; created and maintained detailed computer databases cataloging all in-house seismic and well data of 20+ SE Asian countries.

EDUCATION

M.S., Geosciences, 2003

THE UNIVERSITY OF TEXAS AT DALLAS, Richardson, Texas

GPA: 3.85/4.00

Thesis: concentration in Structural Geology/Tectonics; Title: *Polyphase deformation of the eastern Central province, Oklahoma Ouachita Mountains: evidence in support of episodic orogenesis.*

Activities and Societies: GeoClub: President, 1999-2000; Vice President, 1998-1999; wrote Department newsletter.

M.S., Geographic Information Sciences, 2002

THE UNIVERSITY OF TEXAS AT DALLAS, Richardson, Texas

GPA: 3.85/4.00

Graduate Certification in Geographic Information Systems: 2000

B.S., Geosciences, 1998

THE UNIVERSITY OF TEXAS AT DALLAS, Richardson, Texas

Graduated Cum Laude honors

GPA: 3.68/4.00

Activities and Societies: Golden Key National Honor Society; National Dean's List; Phi Theta Kappa International Honor Society; Vice-President's Honor Roll; Outstanding College Students of America; Pi Kappa Alpha Fraternity

CERTIFICATIONS | PROFESSIONAL AFFILIATIONS

Licensed Professional Geoscientist, No. 15612, Issued 2024

Texas Board of Professional Geoscientists

Certified Professional Earth Scientist, No. 3530, Issued 2018

Society of Independent Professional Earth Scientists (SIPES)

Certified Professional Geologist, No. 11773, Issued 2015

American Institute of Professional Geologists (AIPG)

Certified Petroleum Geologist, No. 6117, Issued 2012

American Association of Petroleum Geologists (AAPG), Division of Professional Affairs (DPG)

Graduate Certificate in Geographic Information Systems, Issued 2000

The University of Texas at Dallas (UTD), School of Economic, Political, and Policy Sciences (EPPS)

Affiliations | Memberships

Sigma Xi, The Scientific Research Honor Society:	Geological Society of America (GSA)
UT Dallas Chapter Board Member and Treasurer	Dallas Geological Society (DGS)
Society of Independent Professional Earth Scientists (SIPES):	Permian Basin Section of The Society for Sedimentary Geology (PBS-SEPM)
Dallas Chapter Board Member and Membership Chair	South Texas Geological Society (STGS)
American Association for the Advancement of Science (AAAS)	West Texas Geological Society (WTGS)
American Association of Petroleum Geologists (AAPG)	

HONORS & AWARDS

Marquis Who's Who	2021
Marquis Who's Who in America	
Nominated and inducted as a biographical listee in this invitation-only publication dating back to 1898.	
The University of Texas at Dallas	2013
Green and Orange Award for Alumni Service	
Given annually to the alumnus whose activities and efforts in support of the University are deemed of significant impact to the University's growth and development.	
Pioneer Natural Resources	2005
Pat on the Back Award	
Corporate level award for work performed on a large acquisition evaluation.	
Martin Marietta Materials	2004
Extra Mile Award	
Corporate level award for cost savings initiatives and efficiency enhancements associated with bringing all GIS data and mapping in-house.	
The University of Texas at Dallas	2000
Teaching Recognition Award for Outstanding Contributions to Education	
Awarded by the School of Natural Sciences and Mathematics to the professor and teaching assistant receiving the most nominations by the student body for teaching excellence and student impact (nominated for same in 1999).	

PUBLICATIONS | PRESENTATIONS

Donegan, Colleen, Williamson, David B. (2022, January). <i>Geosteering: Principles, Implementation, and Best Practices</i> . Professional short course created and taught companywide to Scout Energy, Dallas, TX.	
Williamson, David B. (2021, October). <i>Musings from an Endangered Species: Industry Job Outlook, Role of Geoscience, and Need for Consultants</i> . Presentation given at the meeting of the Petroleum Engineer's Club of Dallas, Dallas Petroleum Club, Dallas, TX.	
Hiss, Kevin A., Williamson, David B. (2020, March). <i>Facies and Stratigraphy of the San Andres Formation (Mid-Permian) Petroleum Province, Northwest Shelf of the Permian Basin, West Texas: A Resurgent Play</i> . Paper presented at the meeting of the South-Central Section of the Geological Society of America, Fort Worth, TX.	
Williamson, David B. , Kimme, Keith, Smith, Charlie H., and Kennedy, Tiffani (2019, October). <i>Horizontal Reservoir Value Increased by Enhanced Geological Knowledge</i> . Poster presented at the meeting of the Eastern Section of the American Association of Petroleum Geologists, Columbus, OH.	
Williamson, David B. , Kimme, Keith, Smith, Charlie H., and Kennedy, Tiffani (2019, October). <i>Horizontal Reservoir Value Increased by Enhanced Geological Knowledge</i> . Paper presented at the meeting of the Mid-Continent Section of the American Association of Petroleum Geologists, Wichita, KS.	

Williamson, David B., Kimme, Keith, Smith, Charlie H., and Kennedy, Tiffani (2019, September). *Horizontal Reservoir Value Increased by Enhanced Geological Knowledge*. Paper presented at the meeting of the Rocky Mountain Section of the American Association of Petroleum Geologists, Cheyenne, WY.

Williamson, David B., Smith, Charlie H., Kennedy, Tiffani, and McKinney, Joe (2019, April). *Geologically Focused Data Review Accurately Characterizes Production Factors in Mature Reservoir*. Poster presented at the meeting of the Southwest Section of the American Association of Petroleum Geologists, Dallas, TX.

Druskoff, Mark and Askren, Hana. (2017, March). *Upside Surprise: Oil Superstar Permian Keeps Delivering*. Forbes. Retrieved from [\[LINK TO FORBES ARTICLE\]](#). (NOTE: Did not author this article but am extensively quoted throughout.)

Illich, Harold, Ferworn, Kevin, Nilson, Gary, Hernandez, Frank, and **Williamson, David B.** (2013, August). *Barnett Gas Geochemistry and Thermal Maturity, Fort Worth Basin*. Paper presented at the meeting of the Unconventional Resources Technology Conference, Denver, CO.

Illich, Harold, Ferworn, Kevin, Nilson, Gary, Hernandez, Frank, and **Williamson, David B.** (2013, April). *Barnett Gas Isotope Geochemistry and Thermal Maturity*. Paper presented at the meeting of the Southwest Section of the American Association of Petroleum Geologists, Fredericksburg, TX.

Williamson, David B. *Polyphase deformation of the eastern Central province, Oklahoma Ouachita Mountains: evidence in support of episodic orogenesis*. Thesis. The University of Texas at Dallas, 2003. [\[LINK TO THESIS PDF\]](#)

Nielsen, Kent C. and **Williamson, David B.** (2003, March). *Episodic orogenesis in the Ouachita Fold and Thrust Belt*. Paper presented at the joint annual meeting of the South-Central and Southeastern Sections of the Geological Society of America, Memphis, TN.

Williamson, David B. and Nielsen, Kent C. (2001, May). *Nanichito Thrust: deformed unconformity in the Central zone of the Ouachita Mountains*. Paper presented at the joint annual meeting of the Rocky Mountain and South-Central Sections of the Geological Society of America, Albuquerque, NM.